

DATA STRUCTURES AND PROCESS MANAGEMENT AND OPERATING SYSTEM

DATA STRUCTURE:

Data structures help in organising and managing of data. They are building blocks for algorithm and programs. Data structure is a way of storing and organising data so that it can be used effectively.

COMMON OPERATIONS:

1. Insertion – Adding a new element.
2. Deletion – Removing an existing element.
3. Searching – Finding a particular element.
4. Traversal – Accessing each element in a data structure.
5. Updating – Changing the value of an existing element.

TYPES OF DATA STRUCTURE:

- Array

An array is a collection of elements stored in consecutive memory locations. Usually, all elements in an array are of the same data type.

For example:

Index 0	1	2	3	4
Value 10	20	30	40	50

- Linked List

A linked list consists of nodes connected to each other. Each node contains data and a link to the next node.

Example: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{NULL}$

- Stack

A stack follows the LIFO (Last in First Out) principle. The element that is inserted last is removed first.

Common operations in a stack include:

Push – Adding an element to the stack.

Pop – Removing an element from the stack.

- Queue

A queue follows the FIFO (First in First Out) principle. The element that enters first is removed first.

Example: $\text{Front} \rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow \text{Rear}$

- Tree

A tree is a hierarchical data structure. It consists of nodes connected in a parent-child relationship.

1. File systems
2. Organization structures
3. Family relationships

- Graph

A graph consists of vertices and edges. Vertices represent objects, while edges represent connections between them.

1. Graphs are used in:
2. Social networks
3. Maps
4. Computer networks
5. Routing systems

PROCESS MANAGEMENT:

Process management is the activity of managing the processes running in a computer system. A process is a program that is currently being executed.

A process usually contains important information such as:

- Process ID
- Process state
- Priority
- Memory information
- CPU time

The operating system decides which process should use the CPU. This activity is called process scheduling.

PROCESS SCHEDULING:

Process scheduling helps in

- Proper utilization of the CPU
- Running multiple programs efficiently
- Improving system performance
- Managing the execution of different processes

PROCESS STATES:

A process can move through different states during its execution.

- New: The process is being created.
- Ready: The process is ready to execute and is waiting for CPU allocation.
- Running: The process is currently being executed by the CPU.
- Waiting: The process is waiting for an event or resource.
- Terminated: The process has completed its execution.

A basic flow of process states can be represented as:

New → Ready → Running → Terminated

CONTENT SWITCHING:

Context switching occurs when the CPU changes from one process to another.

During this process, the operating system saves the current state of one process and loads the state of another process. Context switching helps the system perform multitasking.

OPERATING SYSTEMS:

An Operating System, commonly called an OS, is system software that manages computer hardware and software resources.

It acts as an interface between the user and the computer hardware.

Examples of operating systems include Windows, Linux, macOS, and Android.

FUNCTIONS OF OPERATING SYSTEM:

The major functions of an operating system are:

- **Process Management:** The operating system manages processes and controls their execution.
- **Memory Management:** It manages the main memory of the computer and allocates memory to different programs.
- **File Management:** It manages files and directories stored in the system.
- **Device Management:** The operating system controls input and output devices such as keyboards, printers, and storage devices.
- **Security:** It provides security by controlling access to system resources.

OPERATING SYSTEM STRUCTURES:

The basic structure of an operating system can be understood as follows:

User / Applications → System Call Interface → Process Management | Memory Management | File System Management | I/O → Device Management → Hardware

TYPES OF OPERATING SYSTEMS:

- **Batch Operating System:** Jobs are collected and processed in batches.
- **Time-Sharing Operating System:** Multiple users or processes share the CPU for a short period of time.

- Real-Time Operating System: The system provides quick responses within a specific time limit. These operating systems are used in applications where timing is important.

STUDY DESIGN PROCESS:

A proper study and development process helps in understanding a problem and creating an effective solution.

- Step 1: Understand Requirements

The first step is to clearly understand the problem and the requirements.

- Step 2: Break Down the Modules

The complete system or problem can be divided into smaller modules. This makes the task easier to understand and manage.

- Step 3: Design Data Structures

Suitable data structures are selected depending on the type of data and required operations.

For example:

- Arrays for fixed-size data
- Linked lists for dynamic data
- Stacks for LIFO operations
- Queues for FIFO operations
- Step 4: Design Algorithms

An algorithm is a step-by-step procedure used to solve a problem.

The algorithm should be:

- Simple
- Efficient
- Easy to understand
- Suitable for the given problem
- Step 5: Code Implementation

After designing the solution, the algorithm and data structures are implemented using a programming language.

- Step 6: Testing and Debugging

The program is tested to identify errors and check whether it produces the expected output.

Debugging is performed to correct the errors found during testing.

- Step 7: Documentation

Documentation explains the working, design, implementation, and important features of the system or topic.

Good documentation makes the project easier to understand and maintain.

- Step 8: Review and Improvement

The final work is reviewed and improvements are made wherever necessary.

OVERALL PROCESS FLOW:

The complete process can be represented as:

Requirements → Design → Implementation → Testing → Documentation → Deployment

- Requirements: The needs and objectives of the system are identified.
- Design: The structure of the system, data structures, and algorithms are planned.
- Implementation: The planned design is converted into program code.
- Testing: The program is checked for errors and correctness.
- Documentation: The important information about the system is recorded.
- Deployment: The completed system is made available for use.